

Automated Mock Interview Assessment Through Speech Emotion Detection and Semantic Analysis

Dipali Dakhole
Symbiosis Institute of Technology,
Symbiosis International (Deemed
University), Pune, India
dipali.dakhole@sitpune.edu.in

Prabhat Shankar
Symbiosis Institute of Technology,
Symbiosis International (Deemed
University), Pune, India
prabhat.shankar.btech2022@sitpune.edu.in

Kasturi Sangale
Symbiosis Institute of Technology,
Symbiosis International (Deemed
University), Pune, India
kasturi.sangale.btech2022@sitpune.edu.in

Mayank Saha
Symbiosis Institute of Technology, Symbiosis International
(Deemed University), Pune, India
mayank.sahai.btech2022@sitpune.edu.in

Vanisha Agrawal
Symbiosis Institute of Technology, Symbiosis
International (Deemed University), Pune, India
vanisha.agrawal.btech2022 @sitpune.edu.in

Abstract—Rapid advancements in artificial intelligence (AI) and machine learning (ML) are accelerating the shift toward sustainable automation within Industry 4.0 and smart digital ecosystems. In this context, efficient, low-resource AI systems play a crucial role in enabling scalable and environmentally responsible technological solutions. This study introduces Intervalzyze, an AI-powered mock interview evaluation system designed as a lightweight and energy-efficient web application built using Flask. Integrating speech analysis, emotion detection, semantic similarity, and automated feedback generation, the system contributes to green computing practices by optimizing computational workloads while maintaining high performance. Intervalzyze incorporates an enhanced Wav2Vec2 model trained on the TESS dataset for emotion recognition across seven categories fear, anger, disgust, neutral, sad, happy, and pleasant surprise alongside a BERT-based semantic similarity module that evaluates response relevance using cosine similarity. Audio inputs are transcribed via a pre-trained Wav2Vec2ForCTC model, enabling fully automated, low-power processing suitable for scalable Industry 4.0 and robotics process applications. The system generates transcriptions, emotional states, similarity scores, and personalized feedback, supporting sustainable automation in digital training environments. Experimental results demonstrate strong model performance, achieving a 93.94% F1 score, 95.46% precision, and 94.11% accuracy and recall. The proposed solution performed better over baseline K-Nearest Neighbors (kNN) model, and highlights the potential of energy-efficient AI-driven platforms to support smart, automated, and resource-conscious digital systems aligned with the goals of sustainable engineering and automation.

Keywords—Industry 4.0; robotics process automation; speech emotion recognition; mock interview evaluation.

I. INTRODUCTION

Human resource management and recruitment, like many other industries, are being affected significantly by the rapid advancements in ML and AI [1]. One of the areas where a great deal of progress has been made is automated interview grading and application screening. These AI-powered complementary methods to traditional interviewing can offer a deeper, more realistic, and objective assessment of applicants, while the traditional methods may be time-consuming, arbitrary, and not predictable [2]. These enhancements offer chances to automate and standardize the evaluation procedure, guaranteeing uniformity and impartiality in the distribution of feedback. The main objective is to develop a cutting-edge system that can evaluate

simulated interviews by utilizing a deep learning (DL) model for automatic voice detection and an ML model for text similarity scoring.

This paper presents a new AI-powered online tool called Intervalzyze, developed for the purpose of spoken response assessment in mock interview settings. The key contributions are as follows:

- **Multi-Modal Integrated Evaluation Pipeline:** This system provides thorough feedback on spoken responses in simulated interviews through integrating the features of voice emotion recognition, automatic transcription, and semantic similarity scoring.
- **Emotion Identification Using Enhanced Wav2Vec2:** To accurately identify and classify emotional indications in user speech, this paper's model was improved for comprehensibility of interview responses in terms of voice tone and delivery on TESS dataset.
- **The similarity assessment using a BERT-based model for ideal solution similarity in comments with scores provided by the Cosine Similarity (CS) assures the objective completeness and relevance of the content.**
- **The elements of Intervalzyze's Real-time Feedback System, all combined together, produce a responsive pipeline that can provide real-time, data-driven insight on the candidate's emotional state, speech pattern, and quality of content.**

The manuscript is organized as, in session 2, an existing study and gap analysis on text similarity and speech emotion identification are presented. The suggested strategies and how they are being implemented are covered in depth in session 3. The overall results and model insights are presented in session 4. The manuscript concludes with Session 5.

II. LITERATURE REVIEW

The super smart society (Society 5.0) calls for new and faster techniques for voice recognition, emotion recognition, and speech emotion recognition (SER), to maximize human-machine or human-computer contact and collaboration, [3]-[5]. This section examines a number of studies on text similarity and speech emotion recognition using a systematic review.

A. Speech Emotion Recognition

Latest SER models such as Vesper [6], MFGCN [7], and multimodal DL frameworks [8] have improved assessment; however, they are dataset-dependent. Lightweight ViT [9] and SERDNN [10] lack generalization. FunAudioLLM [11] and Attach-SwiNet [12] advanced multimodals designed but have data imbalance. Other models, including gender-aware, CNN-LSTM, and Wav2Vec2/HuBERT-based systems [13]–[15], have imbalanced results across emotions and datasets. Further studies examined metacognition [16], multimodal features [17], and stress detection [18].

B. Text Similarity

Similarity of text makes assessment of transcription relevance possible. Grading based on LSA [19], NLI for stakeholder classification [20], and bilingual similarity using USE-FAISS [21] reduced the efforts but had many shortcomings in fine-grained and contextual accuracy. Cosine Similarity applied to TF-IDF [22] performed better for title matching compared to the Jaccard approach, whereas BERT- and MPNet-based transformer models improved semantic representation [23]. AI models such as Wav2Vec2 and BERT jointly convert speech-to-text, emotion detection, and semantic similarity. The proposed Flask-based system, Intervalyze, integrates these modules for scalable and accurate mock interview assessment.

III. METHODS AND METHODOLOGY

The multimodal system evaluates mock interviews by integrating SER, transcription, and semantic similarity (Fig. 1). Wav2Vec2.0 extracts embeddings from raw audio and classifies seven emotions using a dense layer [24]. Wav2Vec2ForCTC converts audio to text for reference comparison [25], and a pre-trained BERT model computes CS between transcriptions and reference answers [26].

A. Dataset, Data Preprocessing and Feature Extraction

The TESS dataset contains 2800 WAV files labelled with seven emotions: fear, anger, disgust, neutral, sad, pleasant surprise, and happy. Audio, originally recorded at 24.414 kHz, was resampled to 16 kHz using Librosa, normalized to float32, and standardized as $[-1, 1]$ for Wav2Vec2.0 compatibility (Fig. 2). Emotion labels were mapped to integers, and the dataset was split into 2240 training and 560 testing samples. Pre-trained models were used for transcription and text similarity without additional fine-tuning.

B. Wav2Vec2.0

Wav2Vec2.0 is a self-supervised framework that learns speech representations directly from raw audio without manual annotations. It uses a multi-layer convolutional encoder to extract latent features, a transformer-based context network to model temporal dependencies, and a quantization module for contrastive learning. The mathematical formulation is described in [26]. The input audio signal

$$x = [x_1, \dots, x_T] \in R^T$$

is first processed through a convolutional feature encoder to obtain latent representations.

$$z = f(x) = [z_1, \dots, z_{T'}] \in R^{T' \times d}$$

A transformer encoder then maps these latent to context representations

$$c = g(z) = [c_1, \dots, c_T], \quad c \in R^{d'}$$

The latent vectors are discretized using a codebook $E = \{e_1, \dots, e_k\}$ as

$$q(z_t) = c_k, \quad k = \arg \max_j \text{sim}(z_t, e_j)$$

where $\text{sim}(\cdot, \cdot)$ is CS. The model is trained with a contrastive loss to identify the correct quantized vector among N distractors:

$$L_{\text{contrast}} = \log \frac{\exp(\text{sim}(c_t, q(z_t))/k)}{\sum_i \exp\left(\frac{\text{sim}(c_t, q_i)}{k}\right)},$$

and a diversity loss to encourage full codebook utilization:

$$L_{\text{contrast}} = \frac{G}{\log K} \sum_{g=1}^G H(P_g), \quad H(P_g) = - \sum_i P_i \log P_i$$

The total pretraining loss is

$$L = L_{\text{contrast}} + \alpha L_{\text{diversity}}$$

A custom SpeechEmotionDataset class is created to load TESS audio files, pre-process them with the Wav2Vec2 processor, and map labels to integers. Trainer API from Transformers was used with training arguments to manage the training process. The model was trained on the training set, and validation was performed on the test set after each epoch. The sample audio processing using Wav2Vec2.0 is shown in Fig. 3.

C. Audio to Text Transcription Using Wav2Vec2ForCTC

The pre-trained Wav2Vec2ForCTC model from Hugging Face was used for transcription due to its strong ASR performance. It employs CTC loss with a linear layer mapping hidden states to vocabulary logits for decoding audio. No fine-tuning was performed, as the model generalizes well to diverse speech inputs. Audio (16kHz, float32) was processed and transcribed, and outputs were saved for similarity analysis.

D. CS

The CS algorithm measures similarity between two texts by computing the angular cosine between their vector representations. Sentences are mapped into word vectors, and the similarity is calculated using Equation 1, where a_i and b_i are vector components. CS values range from 0 to 1.

$$\cos \theta = \frac{\sum_{i=1}^n a_i b_i}{\sqrt{\sum_{i=1}^n a_i^2} \sqrt{\sum_{i=1}^n b_i^2}} \quad (1)$$

This study used vector weighting with BERT, a pre-trained transformer by Google, generates contextualized word embeddings that vary with context instead of using fixed vectors.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

The Wav2Vec2.0 model is tuned with hyperparameters, as shown in Table I. Several epochs were taken to analyse the model's performance in terms of accuracy, recall, precision, and recall.

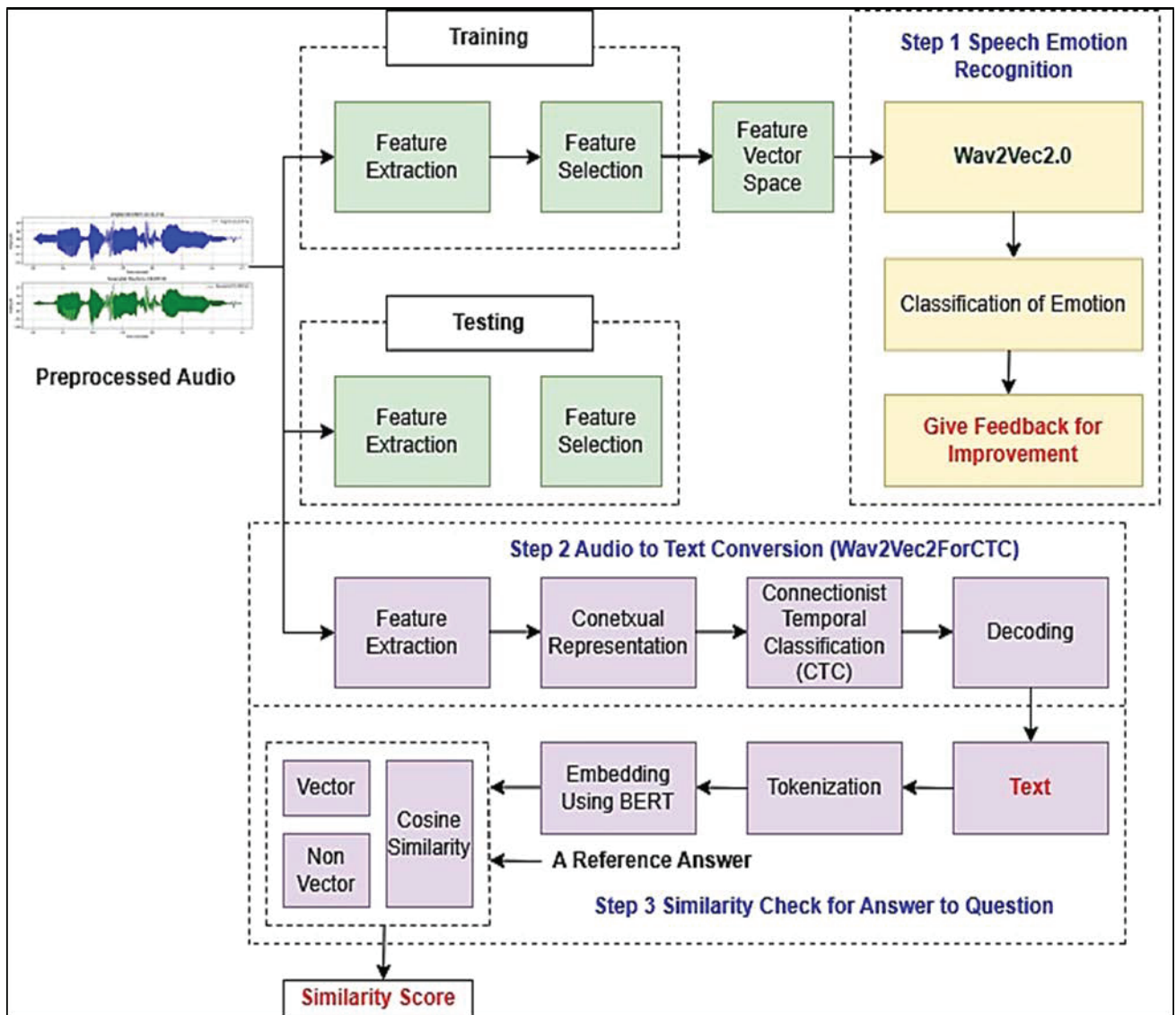


Fig. 1. The Proposed Multimodal System for Mock Interview Assessment using Speech Emotion Recognition and Text Similarity.

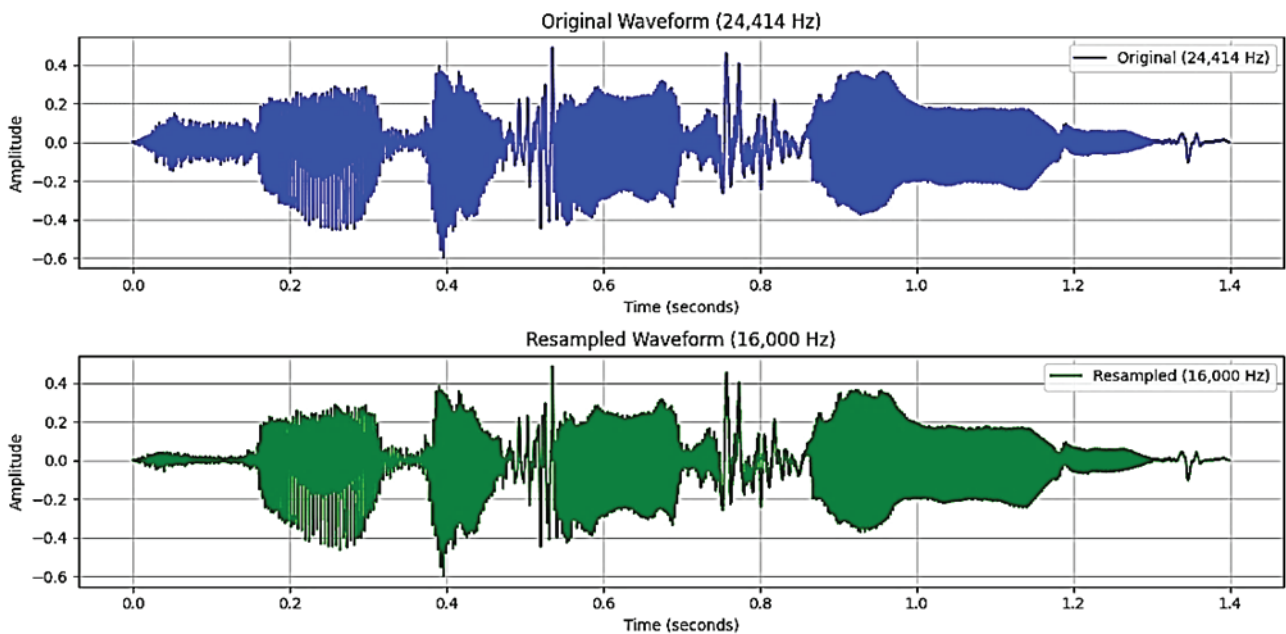


Fig. 2. Sampling of Input Audio Signal.

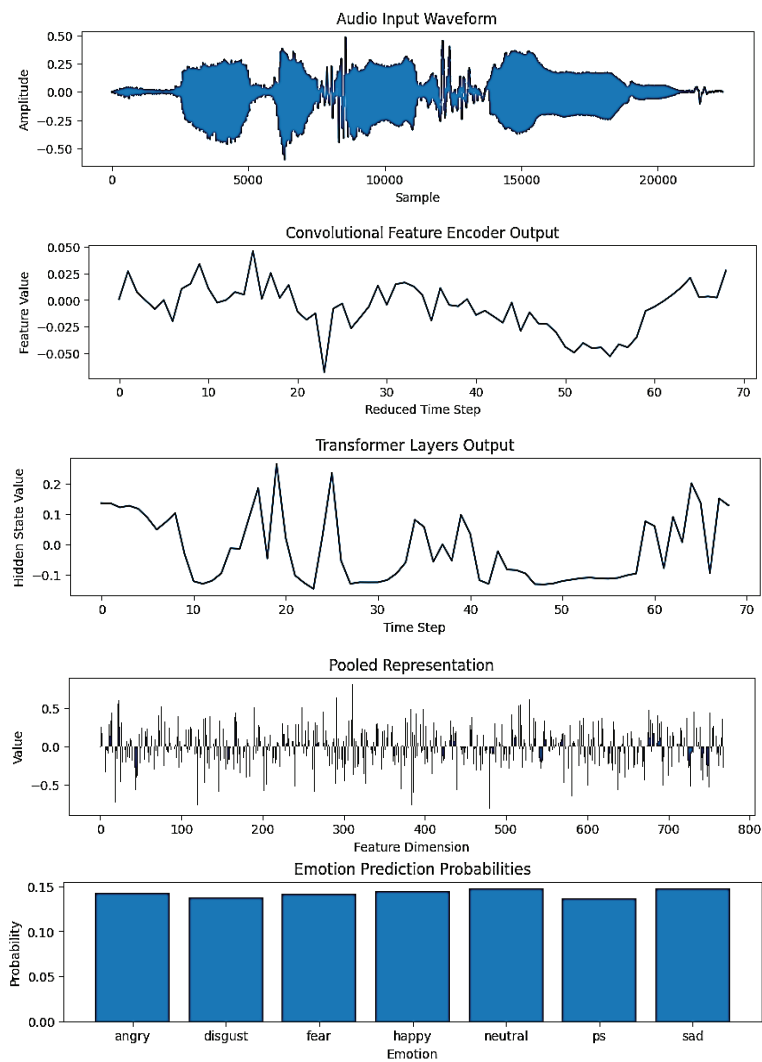


Fig. 3. Sample audio processing using Wav2Vec2.0.

2VEC2.0 HYPERPARAMETER DETAILS

No	Parameter	Value
1	Learning Rate	0.0001
2	Number of Train Epochs	3
3	Train Batch Size	8
4	Test Batch Size	8
5	Weight Decay	2×10^{-2}
6	Mixed Precision Training (FP16)	TRUE

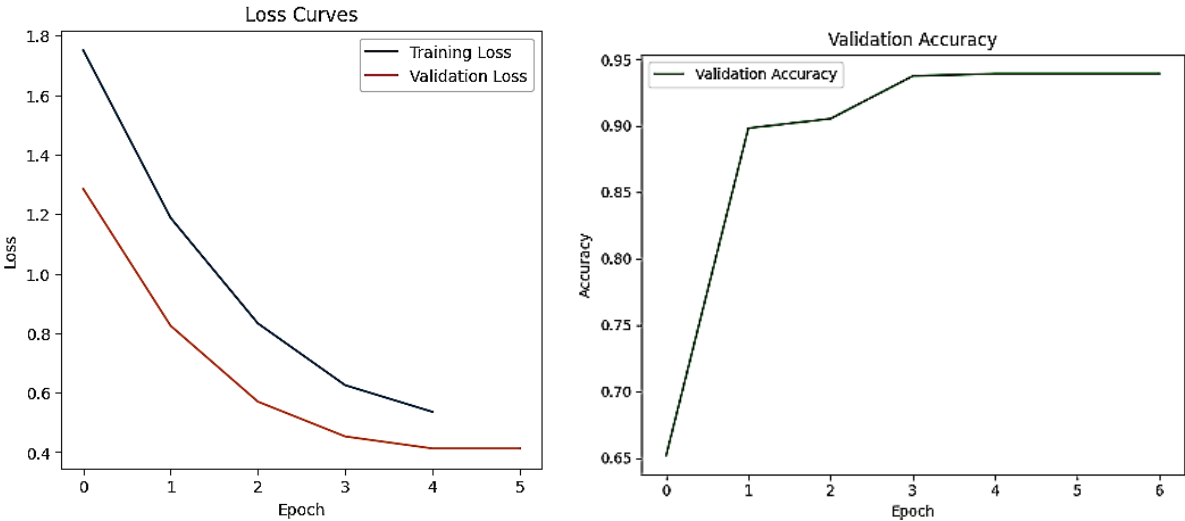
Firstly, the basic training effects of the model are illustrated in Fig. 4a. The loss of the model decreases throughout the process with the increase of epochs for the training set and continues to decline, with the trend of getting closer to zero. In contrast, the loss for the test set tends towards stability after the epochs, reaching approximately 5. The probability of overfitting is reduced as much as possible, and the parameter settings are appropriate, so the whole training effect is relatively ideal according to the loss. Fig. 4b shows that the model accuracy is stabilized after epoch 5.

For the majority of emotions, precision stays high, reaching 100% for anger, disgust, happiness, neutrality, and sadness, indicating minimal rates of false positives. Due to overlapping features, the pleasant surprise emotion has a lower precision of 76.9%, suggesting possible misunderstanding with other emotions. The trend is further

ideal scores, whereas pleasant surprise and happy surprise have the lowest F1 ratings (82.35% and 86.96%, respectively). These outcomes are consistent with findings from prior studies that emphasize how challenging it is to distinguish between emotions that have similar auditory patterns. Notwithstanding these difficulties, Table II demonstrates that the model has balanced precision, recall, and F1 scores, indicating excellent reliability for the majority of categories. The model's total precision of 95.46% suggests a low FP rate. Furthermore, a well-balanced model in terms of FP and FN is indicated by the overall F1 score of 93.94%, which is marginally lower than both accuracy and recall. From the confusion matrix illustrated in Fig. 5, the recognition effect could be observed more clearly by analysing part of the misclassification situations. Negative emotions like disgust are likely to be classified into anger emotion with 1 sample. In contrast, in one instance, the depressing emotion can be categorized as fear. Similar mixed situations are found in the 32 examples where joyful surprise and sad surprise were misclassified, most likely as a result of the pluralistic definition of surprise depending on the positive or negative context of a particular speech.

PERFORMANCE METRICS FOR WAV2VEC2.0 MODEL

Epoch	Training Loss	Validation Loss	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
1	1.7503	1.2844	69.29	69.67	69.29	60.92
2	1.1874	0.8248	84.64	91.78	84.64	80.15
3	0.8327	0.5697	85.54	92.81	85.54	81.08
4	0.6250	0.4527	95.54	96.30	95.54	95.44
5	0.5354	0.4124	94.11	95.46	94.11	93.94



(a) Model Loss for Training and Testing Set. (b) Model Accuracy on Validation.

Fig. 4. Model Loss and Accuracy Curves.

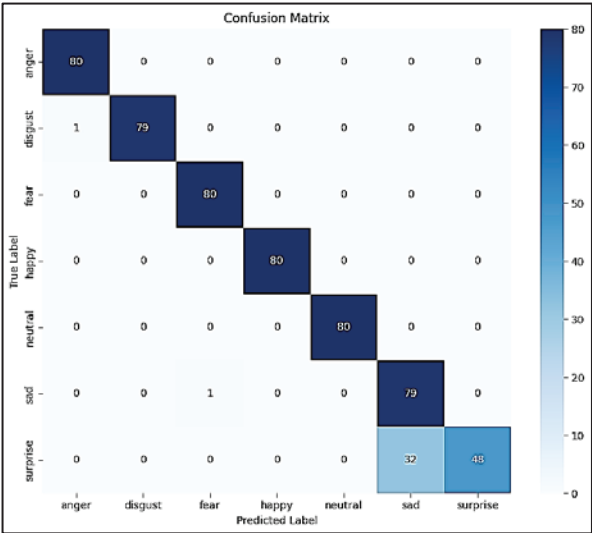


Fig. 5. Confusion Matrix of Results.

COMPARATIVE ANALYSIS OF PROPOSED MODEL WITH BASELINE KNN MODEL

Feature/Metric	Baseline KNN Model	Proposed Model
Model Architecture	Traditional ML (k-NN)	Deep Learning (Transformer)
Input Features	Hand-crafted (Chroma)	Learned (Raw Audio)
Feature Focus	Pitch / Harmony	Timbre / Intonation / Context
Accuracy	71.79%	94.11%
Precision	73.00%	95.46%
F1-Score	72.00%	93.94%

To rigorously validate the performance of the Intervalyze application, a traditional ML k-NN classifier ($k=5$) is trained on Chroma features. It achieved an accuracy of 71.79% and an F1-score of 72.00%, indicating it is insufficient for high-precision recognition (Table III). In contrast, the proposed model, utilizing the fine-tuned Wav2Vec2-xls-r-300m model, achieved an accuracy of 94.11% and an F1-score of 93.94%. Because common emotions like anger, fear, happiness, and neutrality work perfectly under all the right conditions, the speech recognition system's overall performance is acceptable and satisfactory. It shows notable gain performance compared to baseline model.

V. CONCLUSION

Intervalyze: A Mock Interview Evaluator offers a comprehensive and reliable solution for interview evaluation by combining advanced speech analysis techniques like emotion identification, transcription, and semantic similarity assessment. The system provides accurate, context-aware insights into user responses using improved and pre-trained models such as Wav2Vec2 and BERT. Strong performance indicators, such as 93.94% F1 score, 95.46% precision, and 94.11% accuracy, show the model's effectiveness and robustness. The suggested model performed significantly better in terms of accuracy, precision, and F1 score when compared to the baseline kNN model. It offers a strong and accessible solution to enhance interview preparation by offering real-time, personalized feedback through an intuitive Flask web interface.

REFERENCES

- [1] S. Garg, S. Sinha, A. K. Kar, and M. Mani, "A review of machine learning applications in human resource management," *Int. J. Prod.*
- [2] S. Pal, K. Kumari, S. Kadam, and A. Saha, *The AI Revolution*. IARA Publication, 2023.
- [3] D. Mourtzis, J. Angelopoulos, and N. Panopoulos, "The future of the human-machine interface (HMI) in society 5.0," *Future Internet*, vol. 15, no. 5, p. 162, 2023.
- [4] Y. "U. S'onmez and A. Varol, "In-depth investigation of speech emotion recognition studies from past to present: The importance of emotion recognition from speech signal for AI," *Intell. Syst. Appl.*, p. 200351, 2024.
- [5] R. Jahangir, Y. W. Teh, F. Hanif, and G. Mujtaba, "Deep learning approaches for speech emotion recognition: state of the art and research challenges," *Multimedia Tools Appl.*, vol. 80, no. 16, pp. 23745–23812, 2021.
- [6] W. Chen, X. Xing, P. Chen, and X. Xu, "Vesper: A compact and effective pretrained model for speech emotion recognition," *IEEE Trans. Affect. Comput.*, 2024.
- [7] X. Qi, Y. Wen, P. Zhang, and H. Huang, "MFGCN: Multimodal fusion graph convolutional network for speech emotion recognition," *Neurocomputing*, vol. 611, p. 128646, 2025.
- [8] S. Yoheswari, *Advanced Emotion Recognition and Regulation Utilizing Deep Learning Techniques*, 2024.
- [9] S. Akinpelu, S. Viriri, and A. Adegun, "An enhanced speech emotion recognition using vision transformer," *Sci. Rep.*, vol. 14, no. 1, p. 13126, 2024.
- [10] W. Ismaiel, A. Alhalangy, A. O. Mohamed, and A. I. A. Musa, "Deep Learning, Ensemble and Supervised Machine Learning for Arabic Speech Emotion Recognition," *Eng. Technol. Appl. Sci. Res.*, vol. 14, no. 2, pp. 13757–13764, 2024.
- [11] K. An et al., "Funaudiollm: Voice understanding and generation foundation models for natural interaction between humans and LLMs," *arXiv:2407.04051*, 2024.
- [12] T. N. Maghfira, T. Basaruddin, A. A. Krisnadhi, and S. R. R. Pudjiati, "Young-Adult Multimodal Attachment Style Classification," *IEEE Access*, 2024.
- [13] L. Yue, P. Hu, and J. Zhu, "Gender-driven English speech emotion recognition with genetic algorithm," *Biomimetics*, vol. 9, no. 6, p. 360, 2024.
- [14] Q. Ouyang, "Speech emotion detection based on MFCC and CNN-LSTM architecture," *arXiv:2501.10666*, 2025.
- [15] A. Chakhtouna and S. Sekkate, "Unveiling embedded features in Wav2vec2 and HuBERT models for speech emotion recognition," *Procedia Comput. Sci.*, vol. 232, pp. 2560–2569, 2024.
- [16] R. T. Sinvani and H. Fogel-Grinvald, "You better listen to yourself: studying metacognitive efficiency in emotion recognition by voice," *Front. Commun.*, vol. 9, p. 1366597, 2024.
- [17] U. Bhattacharya et al., "Text2gestures: A transformer-based network for generating emotive body gestures for virtual agents," in *Proc. IEEE Virtual Reality and 3D User Interfaces (VR)*, pp. 1–10, 2021.
- [18] S. R. Bandela and T. K. Kumar, "Stressed speech emotion recognition using feature fusion of teager energy operator and MFCC," in *Proc. 8th Int. Conf. Comput., Commun. Netw. Technol. (ICCCNT)*, pp. 1–5, 2017.
- [19] T. Miller, "Essay assessment with latent semantic analysis," *J. Educ. Comput. Res.*, vol. 29, no. 4, pp. 495–512, 2003.
- [20] A. Kuila and S. Sarkar, "From Text to Context: An Entailment Approach for News Stakeholder Classification," in *Proc. 47th Int. ACM SIGIR Conf. Res. Dev. Inf. Retr.*, pp. 2426–2430, 2024.
- [21] L. D. Krisnawati, A. W. Mahastama, S. C. Haw, K. W. Ng, and P. Naveen, "Indonesian-English textual similarity detection using Universal Sentence Encoder (USE) and Facebook AI Similarity Search (FAISS)," *CommIT J.*, vol. 18, no. 2, pp. 183–195, 2024.
- [22] T. P. Rinjeni, A. Indriawan, and N. A. Rakhmawati, "Matching scientific article titles using cosine similarity and Jaccard similarity algorithm," *Procedia Comput. Sci.*, vol. 234, pp. 553–560, 2024.
- [23] Z. He, C. E. Dum Dumaya, and V. Quimno, "Measurement of semantic text similarity," *J. Theor. Appl. Inf. Technol.*, vol. 102, no. 5, 2024.
- [24] F. Li, J. Luo, and W. Xia, "WavFusion: Towards Wav2vec 2.0 Multimodal Speech Emotion Recognition," in *Proc. Int. Conf. Multimedia Modeling*, Singapore: Springer Nature, pp. 325–336, 2025.
- [25] T. Bystrich, "Multilingual automatic phonetic transcription – a linguistic investigation of its performance on German and approaches to improving the state of the art," 2025.
- [26] X. Jin, "Improving paragraph similarity by sentence interaction with BERT," *Expert Syst.*, vol. 42, no. 3, e70003, 2025.